LESSON GUIDE

Technology Plus

Unit 5: Software

3.2.2 – Types of Operating Systems

Lesson Overview:

In this lesson, students will understand the four main types of operating systems, their purposes, and some of their key characteristics.

Students will:

- Describe the key functions and characteristics of desktop, mobile, server, and embedded operating systems.
- Identify operating system types based on their purpose and features.

Guiding Question: What are the key differences between types of operating systems?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 45-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

3.2 – Explain the purpose of operating systems.

- OS types
  - Mobile device
  - Desktop/workstation
  - Server
  - Embedded

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).

Lesson Background

Background Information:

Operating Systems developed early in the history of computing as an easier way for users to interact with the computer and its parts. The original operating systems were what we now call Desktop or Workstation Operating Systems, even though at the time they were only called Operating Systems. Students will likely not need to know individual OS names and specific features; however, they should understand which type different examples fall under.

Materials Needed:

Materials per Class	• Lesson Guide 3.2.2 – Types of Operating Systems • Slides 3.2.2 – Types of Operating Systems • Lab Slides 3.2.2 – OS Explorers
Materials per Group	• Four different colored pens/pencils/markers
Materials per Student	• Student Notes 3.2.2 – Types of Operating Systems • Student Handout 3.2.2 – Types of Operating Systems • Assessment 3.2.2 – Types of Operating Systems (optional)

3.2.2 – Types of Operating Systems

Guiding Question: What are the key differences between types of operating systems?

Lesson Launch (3-5 minutes)

- Ask students if they know how to interact with the computer shown on the intro slide.
- Ask students what an operating system does on a device they use every day – like a phone or laptop.
- Begin by explaining that operating systems manage hardware, support user interaction, and serve as a platform for applications. Over time, they've evolved to support different types of devices and computing needs – from desktop multitasking to mobile responsiveness to server security and embedded efficiency.
- As part of this evolution, operating systems transitioned from 16-bit to 32-bit to 64-bit architectures, allowing them to handle more memory and more complex tasks.

Types of Operating Systems (10-20 minutes)

- Distribute Student Handout 3.2.2 – Types of Operating Systems and encourage them to fill in answers as you go through the lesson.

What Is an Operating System?

- Introduce each operating system type using the slides provided – Desktop, Mobile, Server, and Embedded. Each type includes an overview slide followed by examples to help students understand its purpose and features.
- Students will also review OS classifications like open-source vs. commercial and GUI vs. CLI, which are included in the early slides.

Workstation/Desktop

- Explain that this is the original type of operating system designed for a single machine (most often a desktop/laptop computer) used for general computing tasks.
- Optimized for general computing and multitasking.
- Examples include:
  - **Microsoft Windows**: commercial, largest installed base-OS worldwide
    - Key Versions:
      - Windows 1.0 – 3.x: early GUI layers over MS-DOS (1985)
      - Windows 95: introduction of Start button and Windows Explorer
      - Windows XP: first 64-bit version

        - Windows 8: introduced tile-based Start screen (removed Start button)
        - Windows 10: reintroduced Start button, added Action Center and Cortana
        - Windows 11: redesigned interface with better touchscreen support
    - GUI by default with optional PowerShell CLI
- **macOS**: commercial, UNIX-based OS developed by Apple exclusively for Apple hardware
    - Key Versions:
        - System 1: introduced first widely used GUI (1984); developed internally and based on Xerox and Lisa OS features
        - Mac OS X: first mac OS version based on UNIX
    - Known for stability, GUI consistency, and strong integration with Apple devices
    - GUI by default with built-in Terminal for CLI access
- **Linux Distributions**: open-source, highly customizable
    - Popular Distributions: Ubuntu, Kali, Fedora, Mint
    - Often used by developers, educators, and in systems with older hardware
    - Available in both GUI and CLI-only forms
- **ChromeOS**: commercial, developed by Google exclusively for Chromebooks (laptops) and Chromeboxes (desktops)
    - Key Versions:
        - ChromeOS: originally based solely on Chrome browser (2011)
        - Current versions rebuilt on Gentoo Linux kernel; Google doesn't provide traditional version names
        - ChromeOS Flex: can be installed on older Windows or macOS devices
    - Requires a Google account and consistent internet connection
    - Known for fast boot times and simplicity
    - Popular in K-12 schools due to affordability, automatic updates, and ease of use

Mobile Device

- Explain that the explosion of mobile technology created a demand to design a specific operating system for smartphones, tablets, and other portable devices.
- Optimized for touch interfaces and battery efficiency.
- Examples include:
    - **iOS/iPadOS**: Apple
        - Closed-source, commercial OS
        - Known for strong integration with Apple hardware

- **Android**: Google
    - Open-source core with commercial variants
    - Highly customizable and used across many device brands
- **HarmonyOS**: Huawei
    - Designed for a range of smart devices (phones, TVs, and IoT devices)

Server

- Explain that these are designed to handle multiple concurrent client connections over a network, manage services, and host websites.
- Optimized for scalability, reliability, and security.
- Examples include:
    - Linux-based
        - Open-source options: CentOS, Ubuntu Server, Red Hat Enterprise Linux (RHEL)
        - Despite most Linux workstation operating systems being open-source and free, most Linux-based server operating systems are commercial.
    - Windows-based
        - Commercial and tailored for enterprise use.
        - Includes features like Active Directory and Hyper-V.
    - UNIX-based
        - Used in specialized or legacy enterprise environments.
        - Solaris and AIX
- The table helps students compare the three major types of server operating systems – Linux, Windows, and UNIX – based on key characteristics and typical use cases. Encourage students to focus on what makes each type unique, rather than trying to memorize every detail.
- Use the table to guide a brief discussion about:
- **Licensing models** (open-source vs. commercial)
- **Interface differences** (GUI vs. CLI)
- **Enterprise features and typical environments** (e.g., Windows in schools vs. Linux in cloud servers)
- This comparison reinforces that there's no one-size-fits-all server OS – different systems are optimized for different roles, and understanding their strengths helps in choosing the right tool for the job.

Embedded

- Explain that these are designed to run on specialized hardware that performs dedicated functions – often as part of a larger system, like a smart appliance, car, or industrial controller. These systems are optimized for reliability, efficiency, and limited hardware resources.
- Optimized for efficiency and low-hardware needs.
  - Embedded Linux – open-source and highly customizable, widely used in consumer devices.
  - FreeRTOS – lightweight, open-source RTOS used in microcontrollers and IoT.
  - VxWorks – commercial RTOS used in aerospace, medical devices, and robotics.
  - QNX – commercial, real-time OS used in automotive and industrial systems.
  - Firmware loops – some embedded systems don't use a full OS at all but instead run a single loop of code directly on hardware.

Does It Have an OS?

- This slide is a formative check to help students distinguish between devices that use embedded operating systems and those that don't. Each answer is covered by a white box that reveals the answer and explanation when removed.
- Ask the students to decide if the device runs an embedded operating system. Encourage them to justify their answers using what they have learned about embedded systems.
- After the discussion, click to reveal the answer.

Group the Operating Systems

- Distribute four different colored pens/pencils/markers to groups.
- Ask pairs to create groups for the different examples of operating systems, write a label for their grouping, and circle the examples that fit under that label in the same color.
- Students may ask for more direction, however, consider leaving it open to encourage unique groupings.
- Examples might include:
  - Labeling by Type (Workstation, Mobile, Server, and Embedded)
  - Labeling by Commercial Vendor (Apple products, Windows products, Linux products, etc.)
  - Labeling by Visual vs. Text-Based

GUI vs. CLI: Two Ways to Use an OS

- Use this slide to define and describe each interface type.
- Prompt students to share examples of systems or devices they've used with a GUI.
- Ask if anyone has used a CLI before (besides in the CYBER.ORG Range), even if it was just copying a command from a help article.

When Do We Use Each?

- Walk through the comparison table.
- Highlight that GUIs are common in personal use, while CLIs are powerful tools in professional IT environments.
- Encourage students to think about the pros and cons – not which is "better," but when is each more effective.
- Use this discussion to set the stage for the lab, where students will explore both interfaces directly and reflect on how each handles different tasks.

Lab: 3.2.2 – OS Explorers (15-30 minutes)

- This lab will guide students through several types of operating systems in the Range. Students will perform tasks that highlight top features of the operating system.
- You will need Lab Slides 3.2.2 – OS Explorers. You can present it to the whole class or share the slides with students to work through at their own pace.
- Since students will be working in a few different VMs during the activity, they should use the "Duplicate Tab" trick once all machines have booted.
- Which OS felt the most intuitive or familiar?

    Student answers will vary, but many will likely choose one of the Windows machines, since the layout and navigation are familiar from experience at home or school.

- Which requires the most technical knowledge to use?

    Most students will probably identify Kali Linux, due to its focus on penetration testing tools and its reliance on command-line interfaces for key tasks like password cracking.

- If you had to recommend one for a friend's old laptop, which would you choose and why?

    Students may choose different systems, but Ubuntu Linux is likely a favorite due to its intuitive GUI, pre-installed productivity tools, and ability to run well on lower-spec machines.

- Where did you use CLI vs GUI – and how did that impact your experience?

    Students should recognize that they used CLI most heavily in Kali Linux (e.g., with John the Ripper) and GUI interfaces in Ubuntu, Windows 7, and Server. Reflections will vary – some may enjoy the power and control of CLI, while others prefer the ease of clicking through menus.

Putting It All Together (3-5 minutes)

- Your school is considering switching from Windows laptops to Chromebooks. Based on what you've learned, what are some pros and cons of this move in terms of the operating system?

  Chromebooks use ChromeOS, which is lightweight and cloud-based. A pro is that they load quickly, get automatic updates, and are easier for IT to manage. They're also cheaper. But a con is that they don't support all Windows software, especially programs that require local installs. You'd also need reliable internet since most tasks are done online.

- You're setting up a device to control a smart thermostat. What OS type would be most appropriate and why?

  An embedded operating system would be best because smart thermostats only need to run a few specific functions—like adjusting temperature and connecting to Wi-Fi. Embedded operating systems are optimized for efficiency and don't need a full user interface or extra features.

- Your organization needs to reboot 200 servers for an update. Would you rather use a CLI or GUI tool? Justify your answer?

  A CLI would likely be more efficient because you can write scripts to control multiple servers at once instead of clicking through each one manually. It uses fewer system resources and is faster for repeating tasks, like restarting services or updating software.